

Ishpreet Nagi

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EDUCATION

McMaster University

Bachelor of Applied Science, Computer Science

Hamilton, ON

Sept 2021 — May 2026

- Cumulative GPA: Summa Cum Laude | Dean's Honor List
- Relevant Coursework: Applications of Machine Learning, Natural Language Processing, Data Mining, Data Structures, Algorithms, Databases, Operating Systems, Discrete Mathematics, Probability & Statistics

WORK EXPERIENCE

Software Engineer and Research Coordinator

June 2024 — July 2025

Healthcare Systems Research & Analysis

Waterloo, ON

- Engineered an end-to-end ML pipeline for large-scale geospatial classification using **TensorFlow** and **PyTorch**, benchmarking model variants and shipping the winning architecture at **98% accuracy**
- Built distributed data pipelines in **Python** with **Pandas** and **NumPy** to ingest, clean, and validate multi-million-row datasets, powering training and evaluation workflows and **automating 90% of model deployment**
- Instrumented training and inference with reproducible experiment tracking, GPU-aware batching, and evaluation metrics that surfaced failure modes early and reduced regression cycles
- Directed a team of 4 engineers and coordinated across 8 stakeholders, translating research goals into technical designs and shipping every milestone on time

Machine Learning Analyst and Research Assistant

May 2023 — May 2025

McMaster University - Department of Computing and Software

Hamilton, ON

- Fine-tuned and evaluated deep neural network models under Dr. Douglas Down, building evaluation and model-serving frameworks that achieved **10% accuracy gains** with **15% smaller models**
- Designed a novel **LSTM** architecture with an automated eval loop and **Optuna**-based hyperparameter search, iterating rapidly on transformer-adjacent baselines to **improve effectiveness by 20%**
- Analyzed model behavior using statistical evaluation, ablations, and error analysis to characterize failure modes and guide the next round of architecture and training decisions

PROJECTS

Kollec App (kollec.app)

Sept 2025 — Apr 2026

- Built and deployed a production computer-vision service using a custom **YOLO** segmentation model with embedding-based context awareness and difference hashing, achieving **99% accuracy** across card types and lighting conditions
- Designed a **RAG-style** identification workflow: image embeddings + vector similarity search over a curated card corpus, combined with a lightweight **LLM** reasoning step for disambiguation
- Integrated **OpenAI** and **Claude** APIs for agentic assistance features, applying prompt chaining, tool use, and structured output parsing with careful evaluation of hallucination and failure modes
- Owned the full ML lifecycle end-to-end — dataset curation, training, evaluation, packaging behind a REST API, and monitoring in production — within a team of 7

Technical Executive, DeltaHacks 10 (github.com/deltahacks/landing-10)

Aug 2023 — May 2024

- Instrumented telemetry and behavioral logging to characterize user engagement, using the resulting data to iterate on content strategy and **increase user attention by 25%**
- Shipped the official DeltaHacks 10 platform under a hard deadline to **1000+ applicants** using Astro and TypeScript

Personal Website (ishpreetnagi.com)

July 2024 — Present

- Built a responsive portfolio using **React**, **Astro**, and **Tailwind CSS**, achieving a **95+ Lighthouse score** across all metrics

SKILLS

- **Languages:** Python, C/C++ (familiar), TypeScript, Java, SQL (PostgreSQL, MySQL)

- **ML & DL Frameworks:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Optuna, Hugging Face Transformers (familiar), JAX (familiar)
- **ML Systems:** LSTM/RNN, CNN/YOLO, embeddings, vector search, RAG, prompt chaining, agent workflows, model evaluation & ablation, hyperparameter search, GPU-aware training
- **Data & Infra:** PostgreSQL, distributed data pipelines, Jupyter, Git, Docker, GitHub Actions (CI/CD), Vercel, Supabase, cloud (AWS/GCP — familiar)
- **AI Tooling:** OpenAI API, Claude API, MCP, Cursor, Claude Code, GitHub Copilot, LLM integration, prompt engineering